Task 4

Objectives:

- Learn how to use UART, I2C in HAL level and use MPU6050, BMP280 sensors
- Suggested Course: Ahmed Eldeeb –
- STM website stm32mcu
- Reference Manual Link : Book
- MPU6050 article : ref
- BMP280 article : ref
- Don't forget the YouTube tutorials

Deliverables:

Task1

Using I2C communicate with MPU6050 to get the pitch, roll and yaw angles, then display them on the virtual terminal using UART., what are your notes on the 3 angles?

Task2

Using I2C Communicate with BMP to get the pressure and temperature then display them on the virtual terminal using UART.

Notes:

- All tasks must be implemented in Proteus and Submitted in a .rar file contains the code project and simulation on Proteus.
- Write the whole functions related to Ultrasonic in an external files .h/.c to enhance the readability, also use DOXIGEN doc.